BSC MATHEMATICS & DATA SCIENCE PROJECT INTERIM REPORT

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Contents

Introduction

Introduction to the Problem

In a world where football has grown increasingly and likewise the football market has grown incredibly to the point where billions are spent annually on transfers of players, it is key that data-driven predictions are needed to avoid overpaying and to avoid any future financial difficulties a club may face. It is known that there are many factors in the reasoning behind a player's market value, how much they would be worth, if they were to be sold to another team. The goal of this project is to make a model that can accurately predict the market value of a player based on their age, league, club, statistics and average match ratings across the last five years, across Europe's top five leagues.

Research Questions

Some research questions for this project are: “How accurate is a model at predicting market values using age, position and stats?”, “How influential is a club’s league position in a player’s market value?”, “What is the most influential factor in predicting a player's market value?” These are examples of questions that this project could answer.

Problem Definition

This will be a regression task, where the input is age, encoded position and statistics. The output will be a market value, and show a year-on-year development of a player’s market value.

Relevance to Data Analytics

This project’s relevance to Data Analytics comes similarly to how stock predictions are made. This project shows many plots and insights in the Exploratory Data Analysis file. German based football website “Transfermarkt” is one of the leading football websites when it comes to statistics and player market valuations, making it very useful to this project. In this file there are plots about the last time the file that is being used to gather the market value of many players from Transfermarkt was updated, the biggest risers and fallers from the different seasons selected across all of the top five leagues that have been selected. The sports world has exponentially increased it’s data use to create valuable insights and how to understand how to improve based on the numbers. Football clubs such as Brentford F.C and Brighton & Hove Albion F.C, base their recruitment strategy and academy on data-driven insights, and employ many data analysts.

Core Technology, Research Processes and Architecture Used

In this project, the technologies used are a variety of Python libraries. In the EDA file, Pandas and MatPlotLib are being used to create the insights and visualisations. The ‘kagglehub’ and ‘shutil’ libraries were used to load the data. The market value data itself was found on Kaggle after searching for a dataset that aligned with the goals of this project. The other data was found on the sports website ‘Fotmob’, where they score ratings for players based on their performance from Europe’s top five leagues. This website also provides an average match rating across the entirety of a selected season for each player. ‘FBref’ was also used to collect data from Europe’s top five leagues. This data was from the 2024-25 season, containing valuable club information and statistics. The Fotmob and FBref data was scraped and stored in CSV files to be used in the future.

Hypothesis for a Solution

The expectation of the solution to this project is that the linear regression model trained on the previous seasons’ domestic league stats along with age and position will predict market values with good accuracy. These features chosen show the core performance and demographic factors, which will allow undervalued players from Europe’s top five leagues to be identified.

Literature Review

Literature on football player valuation began as simple econometric models and have grown into complex machine learning applications, the majority being based on performance based metrics such as expected goals and year-on-year differences. This review is focused on literature that is aligned with the research questions, researching how different models predict market value through different factors like age, stats, position, to find gaps in multiple league analysis.

The literature was found from a range of different places. Google Scholar, DkIT EBSCO and arXiv, were the main sources to find the literature. The search for literature was conducted by using different prompts such as, “football data science”, “sport machine learning”, “football prediction model”, “player prediction model”. These queries companied with reading through different literature led to the understanding of what other researchers have done, and how their projects and research were performed.

It was shown throughout the research that the older literature was more focused on a player's performance in their value. Bryson et al. (2012) found that high market valuations were matched with the top performers through goals and assists, however it lacked any predictive power across leagues. Peeters (2018) used international results and basic statistics from those games, against Transfermarkt's crowd-sourced values which Transfermarkt operates itself, and this worked as it forecasted well while ignoring league dependent biases.

As it seemed that more recent literature focused on more in depth technologies, Fernandez et al. (2019) proved this as advanced metrics such as expected goals (xG), creates it's own score based on the scale of zero to one, how valuable a shot is based on many factors. If the score for the xG is extremely high and close to one, the shot is more likely to result in a goal, and this could have not been a goal and completely discarded through a great save or something similar. There was no way for the player who shot be rewarded for almost scoring the goal. It is comparable with the number of goals a player has scored as they may have scored more than their expected goals, creating the image that they could be more clinical, or they could have scored less than their expected goals, making them more wasteful. This metric created a valuable insight into a player's valuation as it was somewhat of a bridge to compare with shots on target or goals. Sulimov (2024) done something quite like Fernandez et al. (2019) through using CatBoost on xG for a possession chain. It was found that through this model, a Mean Absolute Value of €4.26 million in deltas, showcasing xG's edge over simple stats for year to year comparisons. McHale and Holmes (2023) focused on how random forest could be used to predict market value. This was done through using the random forests on metrics such as plus-minus, showing a players influence on a team's performance in a plus for positive influences and minus for negative influence, when a player is on the pitch for their team. This was found to have a Root Mean Square Error of around €10 million, showing that xG normalises league differences across any players.

Literature on different risers and fallers across worldwide football leagues was also found. Herm et al (2014) modelled that there was a 12 percent annual inflation in market values. This was found through using community evaluations of players, to avoid any biases and get a simple understanding of what a player could be worth based on how spectators of a player judge them. Another example found was Veliz (2025) in their Stanford CS224W project where they used a different sort of method for predicting player market values. GraphSAGE was used for their project where they used the example of Kai Havertz transferring from Bayer 04 Leverkusen to Chelsea F.C., then leaving Chelsea F.C. to join Arsenal F.C. This project displayed how the performances he had at these clubs led to the transfer to Chelsea F.C. and then to Arsenal F.C., and how the transfer value was determined based on the performances he had for those clubs.

The most beneficial literature found were from Coxon D. (2023) and Cordeiro L.G. (2023), as these literature used the dataset being planned to use for this project. Reviewing these literature gave a very intriguing perspective on how the data could be used. The information that they were able to display and create insights with showed how it could be used, and where it could be improved. Cordeiro L.G (2023) focuses on correlations between certain statistics and market value of a player, and Coxon D. (2023) specialises in comprehensive EDA, as well as some fundamental modelling. These two pieces of literature easily found on Kaggle, were the most helpful in the idea behind this project and methods to put to use.

Gaps unresolved through researching different literature were quite clear, as there were no Europe's top five leagues comparisons for market predictions, and there were few literature that used free and easily accessible data also. This project will expand and learn from these literature through merging of datasets to create multi-league EDA and regression, which will address biases in age or position.

Exploration of Data

Data Collection

The data needed to obtain the Transfermarkt market values for players was obtained through a Kaggle database. The creator David Cariboo updated the database consistently until nine months ago when he stopped and announced that the database would not be updated again. The average match rating was obtained through the website Fotmob, where it was copy and pasted into CSV files for each separate season across Europe's top five leagues.

Data Description

The Transfermarkt data gave lots of different CSV files, and it took a while to understand what the files would be useful for towards this project. There were seven files downloaded once connecting to the Kagglehub database. These files were "appearances.csv", containing a player identifier, match identifier, club identifier, date of the game, competition identifier, yellow cards, red cards, goals, assists, minutes played. This file at first seemed to be useful but then it was noticeable that the last date recorded on the file was the tenth of April 2025, which meant that not all matches from the 2024-2025 season were recorded. It was found that "club_games.csv" was not going to be useful as it only had the columns game identifier, club identifier, own goals, own position, own manager name, opponent identifier, opponent goals, opponent position, opponent manager name, hosting, is win. These columns were more useful at looking at manager performance and club performance as it did not have any information about a player's performance. "clubs.csv" and "competitions.csv" were also found to be not useful as they just stored information about the different club like the name, squad size, stadium seats, etc. and the competitions stored similar information like the type of competition and country. The "game_events.csv" file caught the eye as it was expected that this file would be able to be used for this project, however upon examining the file the data it stored would not be suitable as the last recorded date was the 29th of March 2025, meaning it did not store all of the data required. The "game_lineups.csv" and "games.csv" files were also irrelevant to the project as they just stored the lineup and match up of teams, with player identifiers but no information of their influence on the game. The most important files from the database were the "players.csv" and "player_valuations.csv" files. These files contained valuable player information, such as the market value, league, age, highest market value, current domestic league identifier, current club name and image url. These files were able to communicate with each other as the identifier matched from the player valuations file resulted in the ability to show a player's market value information, along with their recent valuations to show any potential trends.

The Fotmob collected data is simple understandable data. It has a rank, player name, and average match rating across that season. This data will be easily usable as it will be able to be match with the other data through the name of the player. This file has not been used for EDA as of yet but will be soon.

The club statistics from Europe's top five leagues was also collected. This was copy and pasted from the website FBref.com and stored in the file name "big5_league_stats.csv". This data shows the important stats across all of the 2024-2025 season from all of Europe's top five leagues. Similarly, this data has not been used in the EDA but will be eventually.

Data Cleaning and Exploration

The data needed some marginal cleaning. This was done by using Python. Most of the cleaning required was removing unnecessary columns that are not relevant to the project, and removing players that were not in Europe's top five leagues. The data was explored through using different commands to find the number of players in the data's position, and the country of citizenship of where the different players are from. Through doing this it was found that the most common players position across the data were defenders, and the most common country of citizenship was Spain. It was important for the project to see when the last update of the data was, and it was found that the last major update was in March of 2025, meaning importantly that the data for players valuations was valid up until then. This gave the indication that the project was on track to work appropriately.

Data Visualisation

Visualisations were conducted to explore the data further. Through the use of Python, the historical highest ever market values of players were able to be found, and the image and name of each player was able to be matched with the value. This gave a good understanding of how more recent players were valued higher than players of older seasons, and also to compare with the highest valued players for different positions to see if there is a more “valuable” position group. The biggest risers and fallers from all selected years and leagues were also found. Through this it showed the year on year increase or decrease of a player’s value. This gave the understanding to find what led to the rise or fall of a player's value.

Preliminary Analysis

Preliminary analysis consists of a boxplot of each position group’s market value, and a heatmap showing correlation between age and market value of a player. The boxplot shows that the market value of a player in the “Attack” position group ranges from zero to two hundred million euros. However, there are lots of outliers seen from the boxplot as the outliers for each position are roughly any value greater than twenty five million euros. The heatmap correlation shows that the age vs market value of players who’s market value is greater than one million euro, is -0.12, showing that there is a weak negative relationship between a player’s age and their market value. This means that the older a player is, the more likely the player has a lower market value.

Proposed Future Analysis

The next steps for my project will be to explore average match rating, and club statistics from the chosen seasons. After completing this, the next step will be beginning to create the prediction model. A cleaned player database will be created before creating the prediction model so all necessary information will be able to be obtained. The final parts to do of the project will be to delete unnecessary files, and optimise the functionality of the code to avoid any potential errors.

Code Repository Link

The link to find the progress on the project so far is:

https://gitlab.comp.dkit.ie/D00255640/player-market-prediction

This link shows all merges, includes project meetings and Ethics Application letter, as well as the code and data being used. The journal of meetings can be found in the “docs” folder.

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